MIL-I-38021A(USAF)
28 January 1963
Superseding
MIL-I-38021(USAF)
24 September 1962

MILITARY SPECIFICATION

INDICATOR, LIQUID OXYGEN QUANTITY GMU-39/A

1. SCOPE

1.1 This specification covers one type of liquid oxygen quantity indicator designated Type GMU-39/A. The indicator is of the capacitance type and indicates from 0 to 10 liters of liquid oxygen.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-I-25645	Indicator, Liquid Oxygen Quantity, Capacitance Type, General Specifications for
MIL-L-27160	Lighting, Instrument, Integral, White, General Specification for

STANDARDS

Federal

FED-STD-595	Colors

Military

MS33558	Numerals and Letters, Aircraft Instrument Dial, Standard Form of

(Copies of documents required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 6680

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3. REQUIREMENTS

3.1 GENERAL SPECIFICATION.- The requirements of MIL-I-25645 are applicable as requirements of this specification. When the two specifications conflict, this specification shall govern.

3.2 CONFIGURATION.- The indicator shall consist of a single unit containing all of the equipment necessary to meet the performance requirements specified in MIL-I-25645 and in this specification. The indicator shall be designed so that it may be used either as a master indicator or a repeater indicator, depending on the external wiring.

3.3 CASE.- The length of the case shall not exceed 5.3 inches.

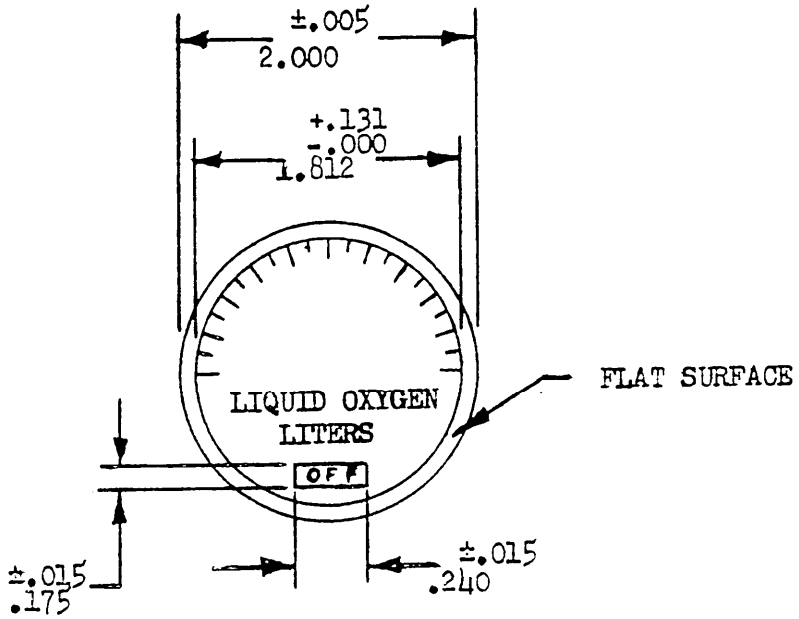
3.4 DIAL.- The dial for the indicator shall conform to figure 1.

3.5 ELECTRICAL CONNECTOR.- The electrical connector shall be as specified in figure 2. The indicator shall be so designed internally that it will function properly, both as a master indicator and as a repeater, when the mating connectors are wired as shown in figures 2 and 3.

3.6 CAPACITANCE.- For design purposes, the capacitance for zero reading shall be 123.5 micromicrofarad (uuf).

3.7 ADJUSTMENTS.- The zero calibration adjustment for the indicator, as a master indicator, shall be capable of adjusting the pointer to zero indication over an input capacitance range of at least 120.0 uuf to 135.0 uuf. With the master indicator adjusted to zero indication at 120.0 uuf input capacitance, it shall be possible to calibrate the master indicator to full scale indication at 178.0 uuf input capacitance by means of the full calibration adjustment. With the master indicator adjusted to zero indication at 135.0 uuf input capacitance, it shall be possible to calibrate the master indicator to full scale indication at 193.0 uuf input capacitance by means of the full calibration adjustment. With the indicator, connected as a repeater, adjusted to zero indication at zero input voltage to pin A, it shall be possible to calibrate the repeater indicator to full scale indication at an input voltage of 6 volts and again at an input voltage of 9 volts applied to pin A.

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NUMERALS AND LETTERS SHALL CONFORM TO MS33558

MARKINGS	HEIGHT OR LENGTH ±.010	WIDTH OF LINE OR GRADUATION ±.005	FINISH COLOR PER FED-STD-595
NUMERALS LIQUID OXYGEN MAJOR GRAD MINOR GRAD SUB MINOR LITERS	.140 .140 .188 .125 .080 .100	.025 .020 .031 .015 .015 .015	37875
BACKGROUND OF DIAL			37038
OFF	.062 .094	.015 .015	37875
OFF FLAG BACKGROUND			31158

DIMENSIONS ARE IN INCHES.

FIGURE 1 - DIAL

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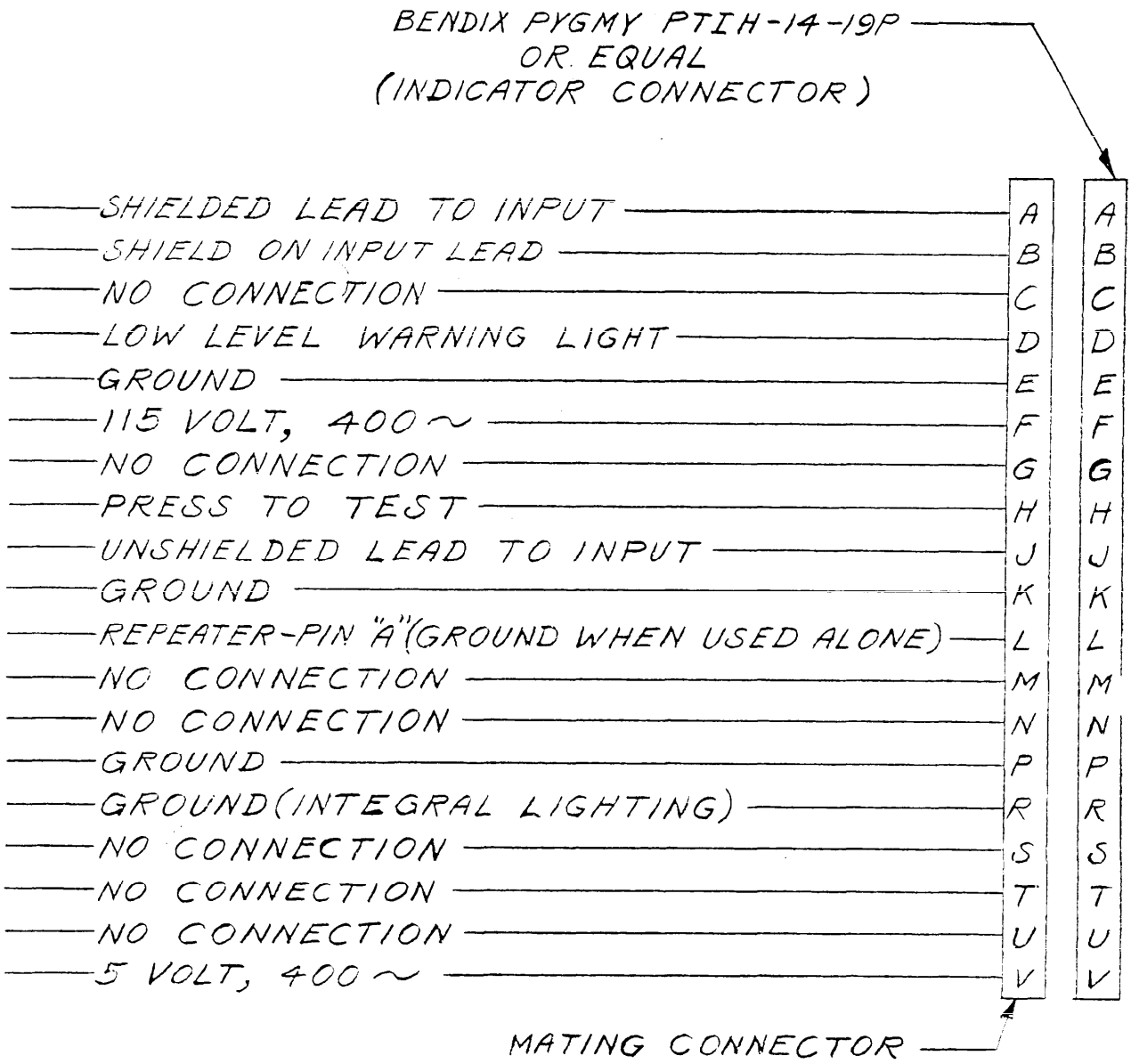


FIGURE 2

4 CONNECTOR WIRING AS MASTER INDICATOR

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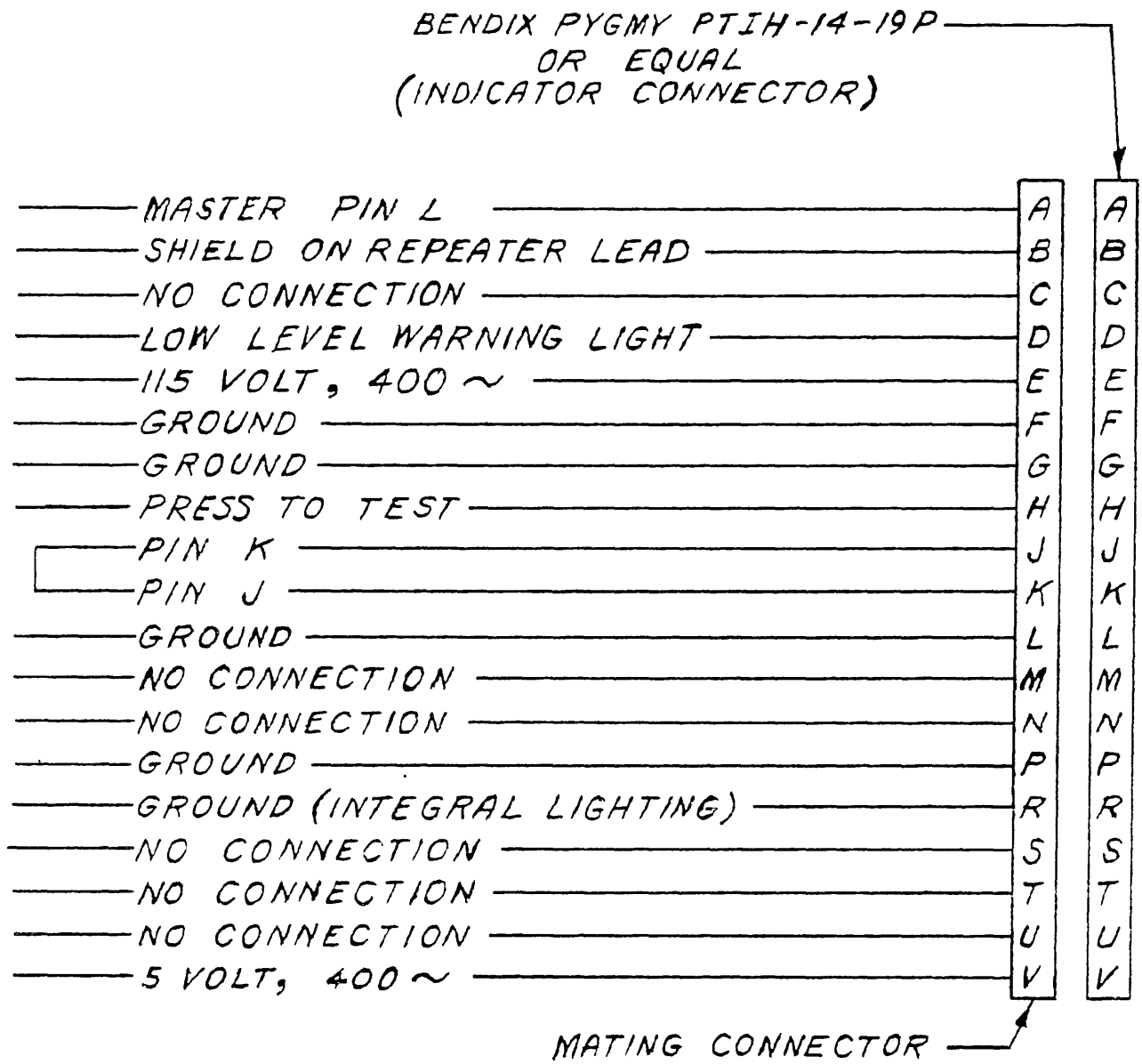


FIGURE 3
CONNECTOR WIRING AS REPEATER INDICATOR

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3.8 WEIGHT.- The indicator shall not weigh more than 1.0 pounds.

3.9 ACCURACY.- When the indicator set is tested in accordance with MIL-I-25645, the accuracy of the indicator shall be in accordance with table I. The tolerance for room temperature readings is the allowable deviation from the nominal calculated values. Repeater tolerance is the allowable deviation from the corresponding master indicator reading. All other tolerances are the allowable deviations from the actual room temperature readings.

TABLE I - TOLERANCE FOR CAPACITANCE READINGS

Test	Tolerance Plus or Minus
Accuracy - room temperature	1.0 uuf
Low temperature operation	1.0 uuf
High temperature operation	1.0 uuf
Vibration	0.6 uuf
Power supply variation	1.2 uuf
Low temperature storage and operation	1.0 uuf
High temperature storage and operation	0.6 uuf
Salt spray	0.6 uuf
Cycling	0.6 uuf
Repeater indication	0.6 uuf

3.10 POWER SUPPLY VARIATION.- Variations of supply voltage from 105 volts to 125 volts and frequencies of 320 to 480 cps shall not cause indicator errors exceeding those specified in table I when the indicator is tested as specified in MIL-I-25645. For the 105-volt test point, the indicator shall be checked at frequencies of 320 and 420 cycles per second. For the 125-volt test point, the indicator shall be checked at frequencies of 320 and 480 cps.

3.11 INTEGRAL LIGHTING.- The indicator shall have integral lighting in accordance with MIL-L-27160.

3.12 POWER FAILURE FLAG.- The indicator shall have a power failure flag that shall be blank for power-on conditions and that shall indicate "OFF" for power-off conditions.

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3.13 ELECTRICAL CHARACTERISTICS AS MASTER INDICATOR.- With the indicator connected as indicated in figure 2 for a master indicator, the impedance from pin L to ground shall be 645K ohms, measured as capacitive reactance. At zero indication, the output voltage from pin L shall be 0 volts, ± 0.2 volts, 400 cps. At full indication, the output voltage from pin L shall be 8 volts, plus 1, minus 2 volts, 400 cps. This output voltage shall vary linearly between zero and full indication. The voltage of pin L of the master shall lead the voltage of pin F of master by approximately 90 degrees with two indicators connected together as master and repeater.

3.14 REPEATER IMPEDANCE.- With the indicator connected as indicated in figure 3 for a repeater indicator, the impedance from pin A to ground shall be 5K ohms.

4. QUALITY ASSURANCE PROVISIONS

4.1 RESPONSIBILITY FOR INSPECTION.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 SAMPLING, INSPECTION AND TESTING.- Unless otherwise specified herein, sampling, inspection and test procedures shall be in accordance with MIL-I-25645, with two indicators connected together as master and repeater. For the purpose of the sampling plan test specified in MIL-I-25645, two indicators, connected as specified above, shall be selected from every 100 indicators or fraction thereof produced.

4.3 INDIVIDUAL TESTS.- The following additional tests shall be conducted on each indicator:

a. Electrical characteristics - master

b. Repeater impedance

c. Repeater calibration adjustment.

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4.3.1 ELECTRICAL CHARACTERISTIC - MASTER.- The indicator shall be connected as indicated in figure 2 for a master indicator with a master test instrument connected to the input leads. Under this condition, a measurement of the impedance from pin L to ground shall be made. This impedance shall be 645K ohms, ± 10 percent, measured as capacitive reactance. Under these same conditions, the voltage output from pin L shall be measured by means of a vacuum tube voltmeter, and shall be 0 volts, ± 0.2 volts, 400 cps. The master test instrument shall then be adjusted to give the dial indications listed in table II with the corresponding output voltage measurements from pin L.

TABLE II DIAL INDICATION AND OUTPUT VOLTAGE

Dial Indication (liters)	Output Voltage (400 cps)
2.5	2.0, plus 0.2, minus 0.5
5.0	4.0, plus 0.5, minus 1.0
7.5	6.0, plus 0.8, minus 1.5
10.0	8.0, plus 1.0, minus 2.0

4.3.2 REPEATER IMPEDANCE.- With the indicator connected as in figure 3 for a repeater, the impedance from pin A to ground shall be measured by means of an impedance bridge. This impedance shall be 5K ohms, ± 10 percent.

4.3.3 REPEATER CALIBRATION ADJUSTMENT.- The indicator shall be connected as indicated in figure 3 for a repeater indicator with a voltage source connected to pin A. This voltage source shall be capable of providing voltages of 0 to 10 volts at 400 cps, with a constant internal impedance of 645K ohms, ± 10 percent. This internal impedance shall be principally capacitive, with a resistive element not to exceed 10 percent of the capacitive reactance. With no voltage applied, the indicator shall be adjusted to zero indication by means of the zero calibration adjustment. Then it shall be determined that the indicator can be adjusted to full indication by means of the full calibration adjustment with 6 volts applied to pin A and again with 9 volts applied to pin A.

5. PREPARATION FOR DELIVERY

5.1 The indicator shall be prepared for delivery in accordance with MIL-I-25645.

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6. NOTES

6.1 INTENDED USE.- The indicator covered by this specification is intended for use in indicating the quantity of liquid oxygen remaining in the converters in F-5A and F-5B airplanes.

6.2 ORDERING DATA.- Procurement documents should specify the following:

a. Title, number, and date of this specification

b. Levels of preparation for delivery required.

NOTICE.- When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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